U92 Quantum Security

Demo lab target, not a real organisation

U92-Vyu free scan: quantum readiness summary

www.demo-bank.example

Every session to www.demo-bank.example uses classical key exchange.



Readiness score

Grade E

High quantum exposure

Post-quantum key exchange

Off

0 of 6 post-quantum groups accepted

Issues to fix

18

4 critical or high, 5 quantum-related

Inside this report

01 **At a glance**
Score, grade and the three fixes that matter most

02 **Exposure and workload**
When recorded traffic could be read, and which teams fix what

03 **The handshake**
Every key exchange offered, and what the server accepted

04 **Findings**
18 issues with what to do, highest severity first

05 **Cryptographic bill of materials**
23 assets, weakest first, plus a CycloneDX file

06 **Verification and next steps**
How to check this report is genuine

Prepared for
Demo Bank (fictional)

Prepared by
U92 Quantum Security, UElement Technologies Private Limited

Report ID
VYU-DEMO-WEB-STD

Scanned
22 October 2026, 14:57 IST

Scan type
Standard public scan

Data must stay confidential for
12 years

UElement Technologies Private Limited, Wakad, Pune, Maharashtra 411057, India. CIN U63990PN2026PTC251047. www.uelement.co, contact@uelement.in, +91 76206 90561.

Based on non-intrusive inspection of public endpoints at the time shown. It is not a penetration test and does not cover internal systems.

At a glance

Every session to www.demo-bank.example uses classical key exchange. Anything recorded today could be read once a large quantum computer exists, and your data needs to stay confidential for 12 years.

Page for leadership

Readiness score
31/100
Grade E: High quantum exposure

Post-quantum key exchange
Off
0 of 6 post-quantum groups accepted

Issues to fix
18
4 critical or high, 5 quantum-related

Cryptographic assets
23
78% quantum-vulnerable or weak

Where you sit on the scale

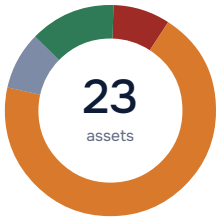
A	B	C	D	E (Your grade)
85+	70-84	55-69	40-54	below 40
Hybrid post-quantum ready	Transitional	Classical, well configured	Classical with gaps	High quantum exposure

Score by area

Key exchange	4	/ 35
Authentication and PKI	12	/ 20
Symmetric ciphers	4	/ 15
Protocol hygiene	8	/ 14
Ecosystem (DNS, mail, web)	1	/ 10

Each bar is the share of that area's points earned. 94% of the scoring weight was tested in this mode.

Your cryptography, by verdict



Weak today	2
Quantum-vulnerable	16
Adequate	2
Quantum-resistant	3

Issues to fix, by severity

1	3	5	9
Critical	High	Medium	Low

Fix these first

1 Critical
Key exchange is classical only: recorded traffic can be decrypted later
CDN / load balancer / web server team · Effort low · F01

2 High
RSA key-transport suites accepted (no forward secrecy)
Web server team · Effort low · F02

3 High
Broken cipher suites accepted
Web server team · Effort low · F03

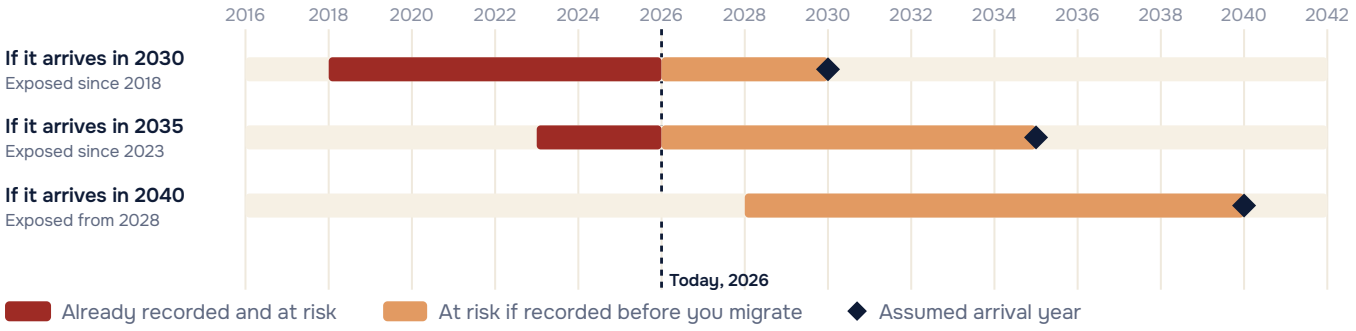
Exposure and workload

When recorded traffic could be read, which areas carry the issues, and which teams will do the fixing.

Page for planning

Harvest now, decrypt later: when could today's traffic be read?

Each row assumes a different arrival year for a quantum computer that can break today's key exchange. With a 12-year confidentiality need, anything recorded on the red and orange stretch is still sensitive when it arrives.

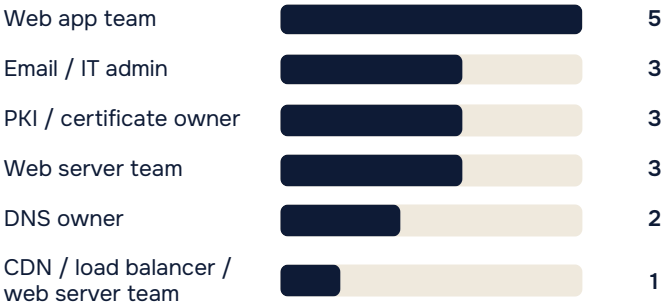


The fix for this chart is a configuration change. Turning on hybrid key exchange (X25519MLKEM768) at your TLS termination point protects every new session from modern browsers. See finding F01.

Issues by area and severity

Area	Critical	High	Medium	Low	Total
Key exchange	1	1	·	·	2
Email	·	1	·	2	3
Ciphers	·	1	·	1	2
Web front end	·	·	3	2	5
Certificates and PKI	·	·	2	2	4
DNS	·	·	·	2	2

Who fixes what



Number of issues each team usually owns. Most fixes are configuration changes, not new products.

Your migration path

1
0 to 3 months
Protect data in transit now
10 findings from this scan
e.g. Key exchange is classical only: recorded traffic can be decrypted late...

2
3 to 9 months
Inventory and crypto-agility
9 findings from this scan
e.g. AES-128 preferred over AES-256

3
9 to 24 months
Move identities and signatures
2 findings from this scan
e.g. Server identity relies on a classical signature (RSA-2048)

4
24 months and on
Retire classical algorithms
0 findings from this scan

The usual order of a post-quantum migration. Phase 1 protects data in transit and is mostly configuration; later phases involve your certificate authorities and partners.

The handshake

We offered www.demo-bank.example:443 every key-exchange group a client can propose, one at a time, and recorded which ones the server accepted.

Page for engineers

21	3	0 of 6
key-exchange groups offered	accepted by the server	post-quantum groups accepted

Post-quantum
ML-KEM, alone or hybrid

- ✕ X25519MLKEM768
- ✕ SecP256r1MLKEM768
- ✕ SecP384r1MLKEM1024
- ✕ MLKEM512
- ✕ MLKEM768
- ✕ MLKEM1024

Classical
broken by a quantum computer

- ✓ x25519 first choice
- ✓ secp256r1
- ✓ secp384r1
- ✕ secp521r1
- ✕ x448
- ✕ brainpoolP256r1tls13
- ✕ brainpoolP384r1tls13
- ✕ brainpoolP512r1tls13
- ✕ ffdhe2048
- ✕ ffdhe3072
- ✕ ffdhe4096
- ✕ ffdhe6144
- ✕ ffdhe8192

Draft Kyber
obsolete pre-standard

- ✕ X25519Kyber768Draft00
- ✕ SecP256r1Kyber768Draft00

Protocol versions

- ✓ TLS 1.3
- ✓ TLS 1.2
- ⊖ TLS 1.1 · VyuAssure
- ⊖ TLS 1.0 · VyuAssure
- ⊖ SSLv3 · VyuAssure

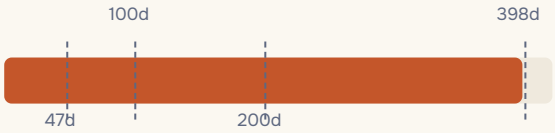
Negotiation

Browser-like offer gets	x25519
Classical listed first gets	x25519
TLS 1.3 ciphers	TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256
Chain validates	Yes
OCSP stapled	No

Certificate served

Subject	www.demo-bank.example
Key	RSA 2048 quantum-vulnerable
Signed with	ECDSA-SHA384
Issuer	Demo Trust Services
Expires in	356 days

Lifetime 397 days against the CA/Browser Forum maximums



Longer than the 200-day maximum that applies to certificates issued since March 2026; the limit falls to 100 days in March 2027 and 47 days in March 2029. Automate renewal now.

Findings

Highest severity first. Each card says what we found, how we know, and what to do. This summary keeps evidence short; U92-VyuAssure reports add full evidence and configuration examples for your stack.

18 to fix

Critical Key exchange is classical only: recorded traffic can be decrypted later

Quantum F01

What we found

Every session to www.demo-bank.example is protected by classical ECDH or RSA key exchange. An adversary who records this traffic today can decrypt it once a cryptographically relevant quantum computer exists (harvest now, decrypt later). With a 12-year data horizon, this is the first thing to fix.

Where: www.demo-bank.example:443

```
TLS 1.3 groups accepted (server's pick order): x2551
9, secp256r1, secp384r1
Offered X25519MLKEM768, SecP256r1MLKEM768, SecP384r1M
LKEM1024 and pure ML-KEM: none accepted
```

What to do

Enable hybrid X25519MLKEM768 key exchange at the TLS termination point (CDN, load balancer or web server). Browsers already offer it, so no client change is needed.

CDN / load balancer / web server team | Effort: Low | Observed

NIST FIPS 203 (ML-KEM); NIST IR 8547 (draft): quantum-vulnerable public-key algorithms deprecated after 2030, disallowed after 2035; NIST SP 1800-38: migration to post-quantum cryptography

High RSA key-transport suites accepted (no forward secrecy)

Quantum F02

What we found

These suites encrypt the session key directly to the server's RSA key. Whoever obtains that key later, through a breach or a quantum computer, can decrypt every recorded session at once.

Where: www.demo-bank.example:443

```
TLS_RSA_WITH_AES_256_GCM_SHA384
TLS_RSA_WITH_AES_128_CBC_SHA
… and 1 more in the full report
```

What to do

Remove all TLS_RSA_WITH_* suites; allow ECDHE (and hybrid ML-KEM in TLS 1.3) only.

Web server team | Effort: Low | Observed

RFC 9325 / BCP 195: recommendations for secure use of TLS

High Broken cipher suites accepted

F03

What we found

The server still accepts suites that are weak without any quantum computer (3DES, RC4, DES, export-grade, NULL or anonymous).

Where: www.demo-bank.example:443

```
TLS_RSA_WITH_3DES_EDE_CBC_SHA
```

What to do

Remove these suites from the server configuration.

Web server team | Effort: Low | Observed

RFC 9325 / BCP 195: recommendations for secure use of TLS

High DKIM key 'selector1' is RSA-1024

F04

What we found

1024-bit RSA DKIM keys are within reach of well-funded attackers today, allowing forged signed mail.

Where: selector1._domainkey.demo-bank.example

What to do

Rotate to a 2048-bit key.

Email / IT admin | Observed

Medium Server identity relies on a classical signature (RSA-2048)

Quantum F05

What we found

Certificates prove the server's identity at connection time, so this is a future forgery risk rather than a harvest-now risk. Public certificate authorities do not issue ML-DSA certificates yet; the work now is inventory, automation and short lifetimes so the switch is routine when they do.

Where: www.demo-bank.example:443

```
Leaf: www.demo-bank.example, RSA-2048, issued by Demo Trust Root R1, Demo Trust Services, SHA-256 f1401c3950ac5624…
```

What to do

Automate issuance and renewal (ACME), keep keys at RSA-2048+/P-256, and track your CA's ML-DSA roadmap.

PKI / certificate owner | Effort: Medium | Observed

NIST FIPS 204 (ML-DSA); NIST IR 8547 (draft): quantum-vulnerable public-key algorithms deprecated after 2030, disallowed after 2035

Medium Identity tokens are signed with classical keys

Quantum F06

What we found

Read from your published JWKS. A future quantum attacker could derive these private keys from the public keys and forge login tokens. Short-lived tokens limit the window; the keys themselves need a migration plan.

Where: https://www.demo-bank.example/oauth2/jwks

```
demo-2026 RSA-2048 (RS256)
demo-ec EC P-256 (ES256)
```

What to do

Keep signing algorithms configurable per key (JWKS 'alg'), shorten token lifetimes, and plan ML-DSA once your identity provider supports it.

Identity / IAM team | Observed

NIST FIPS 204 (ML-DSA)

Medium No HSTS header

F07

What we found

Browsers may still try plain HTTP first, which allows downgrade and cookie theft.

Where: www.demo-bank.example

What to do

Send Strict-Transport-Security with at least a one-year max-age.

Web app team | Observed

Medium Plain HTTP does not redirect to HTTPS F08

What we found

http://www.demo-bank.example answers without sending visitors to HTTPS.

Where: www.demo-bank.example

What to do

Redirect all HTTP requests to HTTPS with a 301.

Web app team · Observed

Medium Outdated JavaScript libraries with known issues F09

What we found

Detected from script URLs on the page (inferred from file names and paths).

Where: www.demo-bank.example

```
jQuery 3.4.1: Below 3.5.0: XSS issues CVE-2020-11022
/ CVE-2020-11023
Bootstrap 4.1.3: Below 3.4.1 / 4.3.1: XSS issue CVE-2
019-8331
```

What to do

Upgrade these libraries.

Web app team · Inferred from public data

Lower-priority findings and what is already in place

	Finding	Severity	What to do
F10	AES-128 preferred over AES-256 Quantum	Low	Put TLS_AES_256_GCM_SHA384 first in the server's TLS 1.3 suite order.
F11	Certificate lifetime exceeds the CA/Browser Forum limit	Low	Automate renewal with ACME so shorter lifetimes are painless.
F12	No CAA record	Low	Publish CAA records naming your CAs.
F13	DNSSEC not enabled	Low	Enable DNSSEC signing (ECDSAP256SHA256) at your DNS provider and publish the DS record at the registrar.
F14	DMARC is in monitor-only mode (p=none)	Low	Move to p=quarantine, then p=reject.
F15	No MTA-STS policy	Low	Publish an MTA-STS policy (start in testing mode) and a TLS-RPT record.
F16	Cookies set without the Secure flag	Low	Add Secure (and HttpOnly, SameSite) to session cookies.
F17	3 third-party scripts load without integrity checks	Low	Add Subresource Integrity (integrity="sha384-...") to static third-party scripts and keep an approved-script inventory.
F18	3 certificates exceed the CA/Browser Forum lifetime limit	Low	Automate issuance with ACME.
F19	Certificate inventory from Certificate Transparency Quantum	Note	Confirm which certificates are in use, retire the rest, and put the remainder under automated management.
F20	No continuous cryptographic inventory (CBOM) yet Quantum	Note	Generate a CycloneDX CBOM for every build and every environment, and review it before release. U92-DeepVyu covers internal discovery.
F21	No security.txt	Note	Publish /.well-known/security.txt with a contact and expiry.

Cryptographic bill of materials

Every algorithm, key, certificate and protocol we found. The same inventory is attached as a CycloneDX 1.7 file, the format CERT-In's BOM guidelines accept.

23 assets

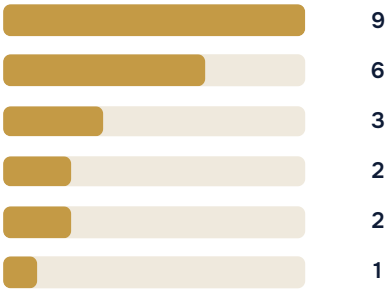
23 assets in total

16 quantum-vulnerable

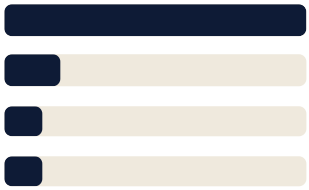
2 weak even today

3 post-quantum or quantum-resistant

By type



By where we found it



Weakest first

Asset	Purpose	Verdict	Evidence
3DES-EDE-CBC	TLS payload encryption	Weak today	Observed
DKIM RSA-1024	Mail signing (DKIM)	Weak today	Observed
TLS 1.3	Transport security	Quantum-vulnerable (Shor)	Observed
TLS 1.2	Transport security	Quantum-vulnerable (Shor)	Observed
x25519	TLS session key establishment	Quantum-vulnerable (Shor)	Observed
secp256r1	TLS session key establishment	Quantum-vulnerable (Shor)	Observed
secp384r1	TLS session key establishment	Quantum-vulnerable (Shor)	Observed
RSA key transport	TLS 1.2 key exchange without forward secrecy	Quantum-vulnerable (Shor)	Observed
RSA-2048	Certificate signature key (leaf)	Quantum-vulnerable (Shor)	Observed
Leaf certificate: www.demo-bank.example	Server authentication (leaf)	Quantum-vulnerable (Shor)	Observed
ECDSA P-384	Certificate signature key (root)	Quantum-vulnerable (Shor)	Observed
Root certificate: Demo Trust Root R1	Server authentication (root)	Quantum-vulnerable (Shor)	Observed

11 more assets are in the attached CycloneDX file.

What U92-VyuAssure adds

This free summary covers the public surface of one host. U92-VyuAssure is the paid, per-scan assessment for organisations that must show evidence to a regulator, auditor or customer.

Pay per scan

Area	This free scan	U92-VyuAssure
Post-quantum key exchange	Main host on its main port	Every subdomain and TLS port, SSH and mail servers, legacy protocol tests
Cryptographic inventory	Summary CBOM	Full CBOM and QBOM to CERT-In minimum elements, crypto-agility score, migration roadmap
Software components	Page scripts with known CVEs	Full SBOM with transitive dependencies, exploit-likelihood ranking, VEX statements you confirm
Code	Public repository, secret counts only	Private repositories, secret locations and git history, CI checks
Exposure	Passive checks only	Owner-approved external vulnerability assessment and performance checks
Regulation evidence	None	Packs for RBI, SEBI, IRDAI, CERT-In, DPDP, GDPR, DORA, PCI DSS, HIPAA and more, with a guided questionnaire
Report	This signed summary	Full signed report, CycloneDX and SPDX files, Excel tracker, re-scan within 30 days

Start at https://www.uelement.co/vyu/#assure or write to contact@uelement.in.

How the score works

Five dimensions with fixed weights: key exchange 35, authentication and PKI 20, symmetric ciphers 15, protocol hygiene 20, ecosystem (DNS, mail, web) 10. Checks that were not run in this mode are removed from both the points earned and the points possible, so the score reflects only what was tested; coverage shows how much of the full weight was tested. Grades: A 85+, B 70-84, C 55-69, D 40-54, E below 40.

Verdicts

- **Quantum-vulnerable:** RSA, elliptic-curve and Diffie-Hellman, broken by Shor's algorithm.
- **Hybrid post-quantum:** classical plus ML-KEM (FIPS 203); safe if either holds.
- **Quantum-resistant:** ML-KEM, ML-DSA, AES-256 and similar.
- **Adequate:** AES-128 and similar, fine under NIST guidance.
- **Weak today:** broken without any quantum computer.

Not covered by this free scan

- Legacy protocol acceptance (TLS 1.0, TLS 1.1, SSLv3)
- Other TLS services on the host (8443, 465, 993, 995, 636)
- SSH key exchange
- Mail server STARTTLS on your MX hosts
- TLS on each subdomain found in Certificate Transparency logs
- Internal services, authenticated APIs and service-to-service mTLS
- Mobile apps, HSMs, key management systems and vendor appliances
- Source code and build pipelines (run the GitHub audit, or U92-DeepVyu for private code)

Verification and next steps

This report and its CBOM are signed by UElement. Check them online, or verify the signature yourself with the public key at https://scan.uelement.co/.well-known/vyu-signing-key.pem.

Report ID	VYU-DEMO-WEB-STD
Generated	22 October 2026, 14:57 IST
Scanner	U92-Vyu 2.1.0, from 127.0.0.1 (demo lab), 19 TLS/SSH connections
Result SHA-256	`5bfa07f73e81954f531e9330e829d95b030529a91186df91c5c4d4a5f047d0b8`
CBOM SHA-256	`636c25fadefafd99c91281ec6f93e6b9fa390134b11f2570feffbc619d57d404`
Signature (Ed25519, key 32db0e74a538ed49)	`KQMPGBOIAEHqTqV+wQvZSVpC3KUzvZm0xYQ75stOvWS5gOYuGdIOQG75neQrfrx8+5Ce236nKwKbXYoRaW9UBA==`
Verify online	https://scan.uelement.co/verify/VYU-DEMO-WEB-STD

What this scan covered

- ✓ **Resolve and vet the target** — www.demo-bank.example → 198.18.10.10
- ✓ **TLS versions, cipher suites and key-exchange groups** — classical key exchange only (x25519)
- ✓ **Certificate chain and keys** — RSA 2048, expires in 356 days
- ✓ **DNS, CAA and DNSSEC** — DNSSEC off, no CAA
- ✓ **Mail authentication (SPF, DMARC, DKIM, MTA-STS)** — 1 MX, 2 DKIM keys found
- ✓ **HTTP headers, page scripts and published signing keys** — HTTP 200, 5 scripts
- ✓ **Certificate Transparency inventory** — 7 live certificates, 7 host names
- 🔒 **Legacy protocols (TLS 1.0/1.1, SSLv3)** — Included in U92-VyuAssure
- 🔒 **Other TLS services on the host** — Included in U92-VyuAssure
- 🔒 **SSH key exchange** — Included in U92-VyuAssure
- 🔒 **Mail server STARTTLS** — Included in U92-VyuAssure
- 🔒 **Subdomain sweep from CT logs** — Included in U92-VyuAssure

Where to go from here

You are here

U92-Vyu free scan

The public surface of one host, at no cost.

Next

U92-VyuAssure

Pay per scan: every endpoint, full SBOM with VEX, owner-approved vulnerability assessment and regulation evidence packs.

Programme

Discovery and migration

U92-DeepVyu for internal systems, Vizor xBOM for continuous inventory, U92-PQc for the move to ML-KEM and ML-DSA.

Talk to the U92 Quantum Security team: contact@uelement.in, +91 76206 90561, www.uelement.co/platforms/u92-quantum/

This assessment inspected only what is publicly reachable, using standard connections and public data sources, at the time shown. Results can change as configurations change. Where a row is marked inferred, confirm it with the system owner before acting on it. Nothing in this report is a guarantee of security or a legal opinion. UElement Technologies Private Limited, Wakad, Pune, Maharashtra 411057, India. CIN U63990PN2026PTC251047.